

Taylor Ely

School of Life Sciences, University of Hawai'i at Mānoa

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<https://taylorely.github.io/>

Education

University of Hawai'i at Mānoa

PhD Candidate in Zoology

Thesis: *Connectivity of Marine Communities in the North Central Pacific*

Advisor: Dr. Peter Marko

Expected January 2027

current GPA: 4.0

University of California, Los Angeles

Bachelors of Science Marine Biology

Minor in Environmental Systems and Society

Senior Honors Thesis: *Investigating Temporal Variation of eDNA in a Marine Environment*

Advisor: Dr. Paul Barber

June 2018

GPA: 3.957

Employment and research experience

Aug 2019 - current

Graduate Research Assistant

Marko Lab, University of Hawai'i at Mānoa, Honolulu, Hawai'i

- Collected, processed, and analyzed eDNA samples
- Developed eDNA population genetics pipelines in R
- Simulated larval dispersal into Hawai'i with python and OceanParcels

Feb 2019 - Jun 2019

Research Technician

Carlson Lab, Bowdoin College, Brunswick, Maine

- Full-time lab manager
- Processed parrotfish gut samples for MiSeq and qPCR
- Created a laboratory and bioinformatic protocol for undergraduates

Jun 2018 - Aug 2018

Oregon Sea Grant Summer Scholar

Department of Fish and Wildlife, Marine Resources Program, Newport, OR

- Monitored sea star wasting syndrome (SSWS)
- Expanded the monitoring program through successful collaborations
- Led a team of 3-7 staff and volunteers during intertidal field sampling

Oct 2015 - Jun 2018

Undergraduate Research Assistant

Barber Lab, University of California Los Angeles, Los Angeles, California

- Collected, processed, and analyzed eDNA samples
- Extracted seagrass DNA for genomic analyzes

Jun 2017 - Aug 2017

Invertebrate Zoology Intern

Meyer Lab, Smithsonian National Museum of Natural History, Washington D.C

- Extracted tissue samples from ARMS across Indonesia
- Prepped samples for Sanger sequencing
- Assembled and analyzed sequence chromatograms using Geneious

May 2015 - May 2016

Public Programs Intern

Santa Monica Pier Aquarium, Santa Monica, CA

- Organized monthly scientific talks
- Created advertisements, social media posts, and displays using Canva

Publications

* = equal authorship

Ely TD, Marko PB. 2026. Resolving emergent patterns in community genetics with environmental DNA. *Molecular Ecology* 35, no. 13: e70432.

Nichols PK, Fumo JT, Lopes Jr KH, Ely TD, Nash EN, Timmers MA, Sherwood AR, and Marko PB. (2026). When cryptogenic species are also cryptic: reframing biogeographic uncertainty in the environmental DNA era. *Biology Letters* 22.4

Nichols PK, Sherwood AR, Fraiola KM, Ely TD, Hauk BB, Counsell CW, Porter ML, Marko PB. (2025). Detection of a Non-Indigenous Marine Macroalga (*Acanthophora spicifera*) With Environmental DNA From Surface Seawater. *Aquaculture, Fish and Fisheries* 5, no. 6: e70135.

Fumo JT, Nichols PK, Ely T, Marko PB, Moran AL, Powell BS, Williams TM, Kosaki RK, Smith CM, Lopes Jr KH, Smith JE. (2025). A predictive framework for identifying source populations of non-native marine macroalgae: *Chondria tumulosa* in the Pacific Ocean. *PeerJ*, 13: p.e19610.

Ely TD*, Mukai GNM*, Kobayashi DR, Marko PB, Moran AL, Wren JLK. (2024). Characterizing key sensitivities for modeling long-distance dispersal in the North Central Pacific Ocean. NOAA Technical Report.

Ely T, Patten N, Naisbett-Jones LC, Spencer ET, Willette DA, Marko PB. (2023). Molecular identification of critically endangered European eels (*Anguilla anguilla*) in U.S. retail outlets. *PEERJ*. 11: e14531

Moran AL, Toh M-WA, Lobert GT, Ely T, Marko PB. (2021). Egg masses and larval development of the Antarctic cephalaspidean slug *Waagelea antarctica*, with notes on egg masses of the related *Antarctophiline alata*. *Journal of Molluscan Studies*. 87(3).

Ely T, Barber PH, Man L, Gold Z. (2021). Short-lived detection of an introduced vertebrate eDNA signal in a nearshore rocky reef environment. *PLOS ONE* 16(6): e0245314.

Blanchette A*, Ely T*, Zeko A*, Sura S, Turba R, Fong P. (2019). Damselfish *Stegastes nigricans* increase algal growth within their territories on shallow coral reefs likely via enhanced nutrient supplies. *Journal of Experimental Marine Biology and Ecology*. 513: 21-26

Grants

2024 - 2026	Jessie D. Kay Memorial Award in Zoology (\$10,000) <i>What are the dispersal pathways for marine species to the Hawaiian Archipelago?</i>
2023 - 2024	Elizabeth Alison Kay Endowed Award (\$2,500) <i>Dispersal patterns of marine organisms in the Hawaiian Archipelago</i>
2020 - 2021	The Charles H. And Margaret B. Edmondson Research Fund (\$2,000) <i>Community-wide genetic patterns revealed through eDNA</i>
2019 - 2024	NSF Graduate Research Fellowship Program (\$118,500) <i>Using eDNA to examine the impact of warming oceans on marine fish communities</i>

Honors & Awards

2025 - 2026	UHM Zoology Graduate Program Achievement Scholarship (\$1,000)
2024 - 2025	Honolulu ARCS Scholar (\$6,000)
2024 - 2025	Environmental DNA (eDNA) Conference Bursary (\$1,500)
2017 - 2018	Graduated Summa Cum Laude and Departmental Highest Honors from UCLA
2017 - 2018	UCLA Undergraduate Research Scholarship Program award (\$5,000)
2017 - 2018	Whitcome Research Undergraduate Summer Fellowship at UCLA (\$3,000)
2015 - 2016	UCLA Promise Award Scholarship (\$2,000)
2014 - 2018	Dean's Honors List at UCLA
2013 - 2014	California State PTA Scholarship (\$500)

Presentations

Talks

Jun 2026	"Emergent Patterns in Community Genetics with environmental DNA". <i>Invited Talk at SSB sponsored symposium</i> . Evolution 2026. Cleveland, OH.
May 2026	"Larval Journals: Where do Hawaiian marine species come from and where do they go?" Albert L. Tester Memorial Symposium. Honolulu, HI.
Apr 2025	"Population Genetics of Marine Species in Hawai'i using eDNA". Albert L. Tester Memorial Symposium. Honolulu, HI. <i>Honorable Mention, Best Graduate Student Talks</i>
Feb 2025	"Analysis of eDNA as a community genetics tool". 2nd Australian & New Zealand Environmental DNA (eDNA) Conference. Wellington, New Zealand.
Apr 2024	"eDNA: a promising tool for population genetics". Albert L. Tester Memorial Symposium. Honolulu, HI.
Apr 2022	"Testing eDNA as a fine-scale population genetics tool in marine ecosystems". Albert L. Tester Memorial Symposium. Honolulu, HI.
May 2017	"Territorial damselfish enhance nutrient supplies to algal turf in shallow coral reefs". Marine Biology Quarter Symposium. Los Angeles, CA.

Posters

Dec 2025	"Marine Community Genetics: A Test in the Hawaiian Islands". 8 th Annual Meeting of the eDNA Society. Yamaguchi, Japan.
Aug 2018	"Monitoring Sea Star Wasting in Oregon's Marine Reserves". Oregon Sea Grant Final Symposium. Newport, OR.
May 2018	"Investigating Temporal Variation of eDNA in a Marine Environment". UCLA's Annual Biology Research Symposium. Los Angeles, CA. <i>3rd place Undergraduate Poster</i>

Teaching

July 2024	Environmental DNA team leader , <i>R/V Thompson</i> Trained and supervised a team of undergraduates and post-baccs to process eDNA samples on a research expedition with ongoing mentoring
Aug 2019 - May 2020	Teaching Assistant , University of Hawai'i at Mānoa Ecology and Evolutionary Biology Lab and Invertebrate Biology

Service

Professional

- 2025 – 2026 **Volunteer** for Albert L. Tester Memorial Symposium, University of Hawai‘i at Mānoa, helped set up symposium, ran AV for sessions, ran timing for sessions
- 2023 - 2026 **Reviewer** for manuscripts submitted to *Marine Ecology Progress Series*, *Molecular Ecology Resources*, *Proceedings of the Royal Society B: Biological Sciences*, *Aquatic Conservation: Marine and Freshwater Ecosystems*
- 2025 - 2026 **Student Social Chair**, School of Life Sciences, University of Hawai‘i at Mānoa
Co-led weekly *pau hana* events to foster community and informal engagement among graduate students and faculty
- Oct 2024 **Panelist**, Navigating the Publishing Landscape Workshop, University of Hawai‘i at Mānoa, Provided a graduate student perspective on academic publishing

Community

- 2024 - 2026 Various workdays on O‘ahu: Paepae o He‘eia, Ka Popo Lo‘i o Kānewai, Waiale‘e Lake Pono, 808 cleanups, Kaiāulu ‘O Kahalu‘u,...
- 2024 - 2025 International Save the Vaquita Day at the Waikīkī Aquarium
Lead volunteer, ran an educational booth
- 2020 - 2023 Various STEM outreach for kids: Kupu Hawai‘i STEM outreach; Career Day at Dreamhouse HS; The Science of the Seuss Library Event
- 2020 - 2021 Science Fair Judge, Leeward Science and Engineering Fair
Volunteer judge for environmental science and biological sciences categories
- 2013 - 2014 Environmental Message of the Month at Palos Verdes Intermediate School
Founded and ran this program to educate students and parents about various topics and provide workdays for the students to engage with the community,
<http://monthlyenvironmentalmessageweebly.com/>

News & Media

- Sep 2024 UH News Article
<https://www.hawaii.edu/news/2024/09/08/research-voyage-marine-links-north-pacific/>

Other Certifications and Training

- AAUS Scientific Diver (since 2016)
- ZOOL 750: An Introduction to place-based methodologies in Hawai‘i
- NREM 691: People and the Sea
- PEPS 662: Systematics and Phylogenetics